DS603 – Data Structures and Algorithms in Python

Course Placement: Data Structures and Algorithms is a core course for first-year students of the MSc Data Science program.

Course format: It is **3 hours of lecture and 2 hours of lab practical** every week.

Prerequisite: None expected

Course Content: Data Structures and Algorithms in Python introduces students to the fundamental concepts and techniques essential for efficient problem-solving in computer science. Through lectures, coding labs, and problem-solving sessions, students will develop a solid foundation in data structures and algorithms, preparing them for more advanced coursework and real-world applications in data science.

Text Books:

- Data Structures and Algorithms in Python by Michael Goodrich, Roberto Tamassia, Michael Goldwasser, Wiley Publication
- Python crash course: a hands-on project based introduction to programming, 2nd Edition, by Eric Matthews, San Francisco No Starch Press 2019.
- Learn Python the hard way: a very simple introduction to the terrifyingly beautiful world of computers and code, 3rd Edition, by Zed Shaw, Upper Saddle River: Addison-Wesley, 2014

Other Resources:

- www.python.org

Assessment method:

- Pop-up Quizzes: 10%
- Lab Assignments: 15%
- Project: 20%
- First In-Semester Exam: 10%
- Second In-Semester Exam: 20%
- End-Semester Exam: 25%

Course Outcomes:
After successful completion of the course,

- Students will learn the Python programming language and tools such as Jupyter Notebook through hands-on exercises.
- Students will be able to solve logical problems by implementing the solutions using the Python programming language.
- Students will be able to store and retrieve data using Python data structures such as lists and dictionaries.
- Students will be able to implement different general data structures, such as linked lists, stacks, queues, and trees and implement CRUD operations on these data structures
- Students will implement different searching and sorting algorithms and perform the time and space complexity analysis.

P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12
X	X	X						X			X

Lecture Schedule:

SI. No.	Description	No. of Lectures
1	Python Basics: Why Python for Analytics?, Features of Python useful for Data Science	1
2	Variables and Simple Data Types: Data Types, Variables, Strings, Numbers, Booleans, Casting, Comments	2
3	Expressions, Operators, Precedence: Arithmetic, Assignment, Comparison, Logical	1
4	Python Data Structures: Lists: What is a List?, Indexing to access the List elements, Changing, Adding and Removing elements, List Slicing, Tuples, Membership operators (in, not in), InstanceOf (is, not is) Tuples and Sets: Unchangeable structures Dictionaries: Accessing or Modifying Values, Adding a Key, Removing Key-Value Pair Nested List or Dictionaries: List of Dictionaries, Dictionary of Lists	4
5	Control Flow: Conditions: if-elif-else, numerical comparison, checking for Multiple Conditions, using if with Lists and Dictionaries Loops: while, for, for..in, Looping through the List, Looping through the Dictionary	4
6	Functions: Library Functions: sorted, zip, reverse, enumerate User Defined Functions: Defining, Passing Arguments, Return Values,	2
7	Classes: Creating Class, Working with Classes and Instances (Objects), Inheritance, Importing Classes	3
8	Files and Exceptions: Reading from a file, Writing to a file, Exception Handling	3
9	Algorithmic analysis: counting operations, asymptotic analysis, big-Oh notation, time- and space-complexity, best-, worst-, and average-case analysis.	4
10	Examples of analysis: Searching: sequential, binary; Sorting: Bubble, Selection, Insertion Sort, Merge sort, Quicksort.	5
11	Linear Data Structures: Stack, Queue, Hash (implementation as an array/ list).	2
12	The concept of abstraction: functional and data."Linked" lists: find, insert, delete; Binary search trees: find, insert, delete; The concept of recursion; efficient evaluation of recursive definitions; Height-balanced binary search trees;	6